

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – INDUSTRY PROBLEM TASK (5 Questions)

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO2 – Precision (BL3)	Assessment:	Assessment Tool 1 – Industry Problem Task
Weightage:	40% of CIA	Total Marks:	50 (5 Questions × 10 Marks)
CO2 Statement:	<i>Apply appropriate visualization methods to represent distributions, proportions, and relationships in data.(BL3)</i>		
Student Name:	Vanumu surya satvika	Reg. No.:	192524254
Section / Batch:	AIDS	Date:	

Code of Conduct

I CH Likhitha (192424239) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Likhitha

Instructions

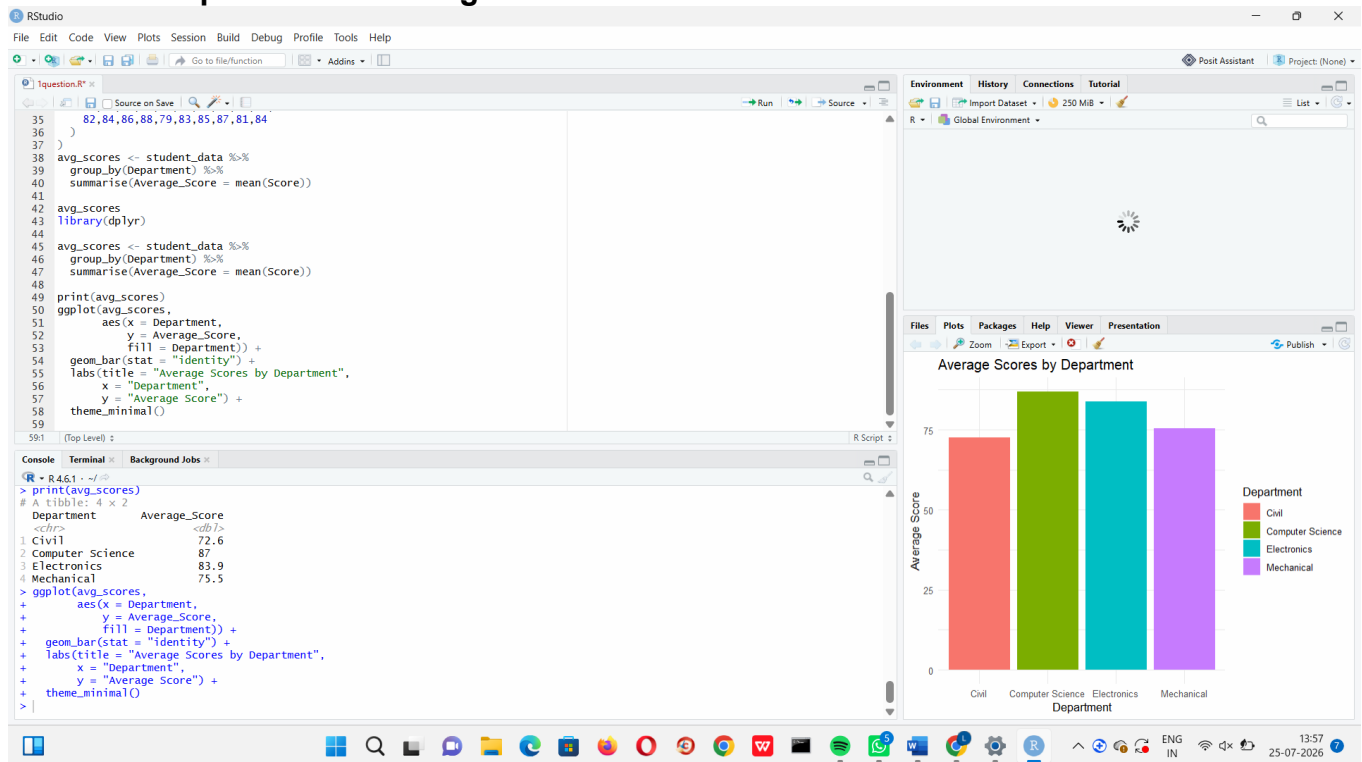
- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

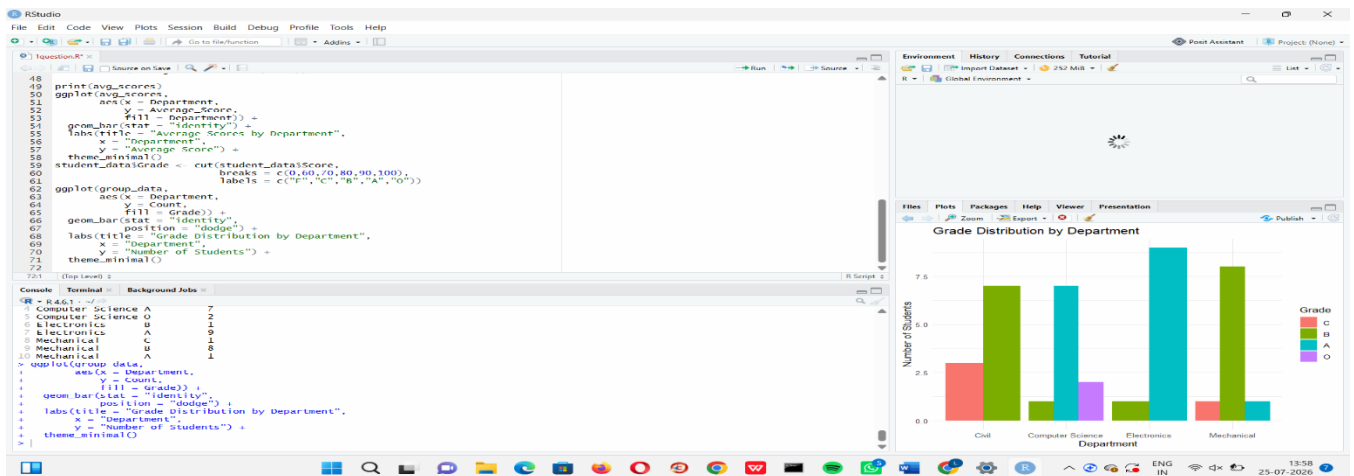
Section / Topic	Focus	Q Nos	Marks
Categorical Data Visualization	Comparing category-wise revenue, proportions, and seasonal trends using bar, grouped, and stacked charts	1	10
Distribution Analysis	Understanding defect distribution and variability using histograms, boxplots, and violin plots	2	10
Relationship Visualization	Exploring relationships between variables using scatter plots, heatmaps, and multivariate techniques	3	10
Geospatial & Temporal Visualization	Analyzing delivery performance using maps, time-series plots, and heatmaps	4	10
Distribution Comparison	Comparing patient data distributions using ridgeline plots, density plots, and boxplots	5	10
GRAND TOTAL:		50 Marks	

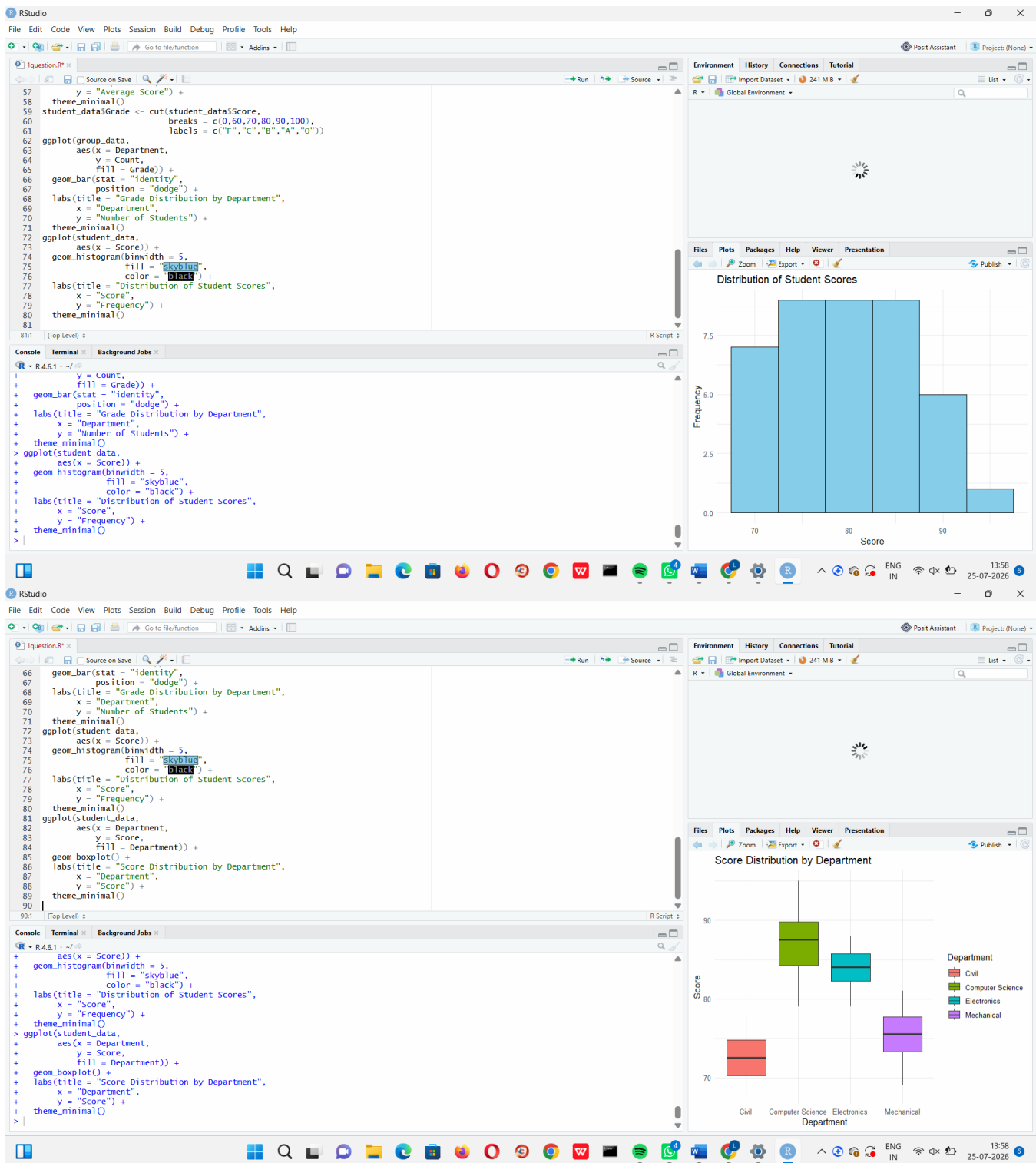
Annexure A – Industry Problem Task

1.A university wants to analyze the academic performance of students across different departments (Computer Science, Mechanical, Civil, Electronics, etc.). The administration needs to compare average scores, identify top-performing departments, and examine variations in student results.

Task: Design suitable visualizations to represent student scores and performance distributions. Justify the use of bar plots, grouped bars, histograms, and boxplots. Explain how these visualizations help identify performance trends and support academic improvement strategies.

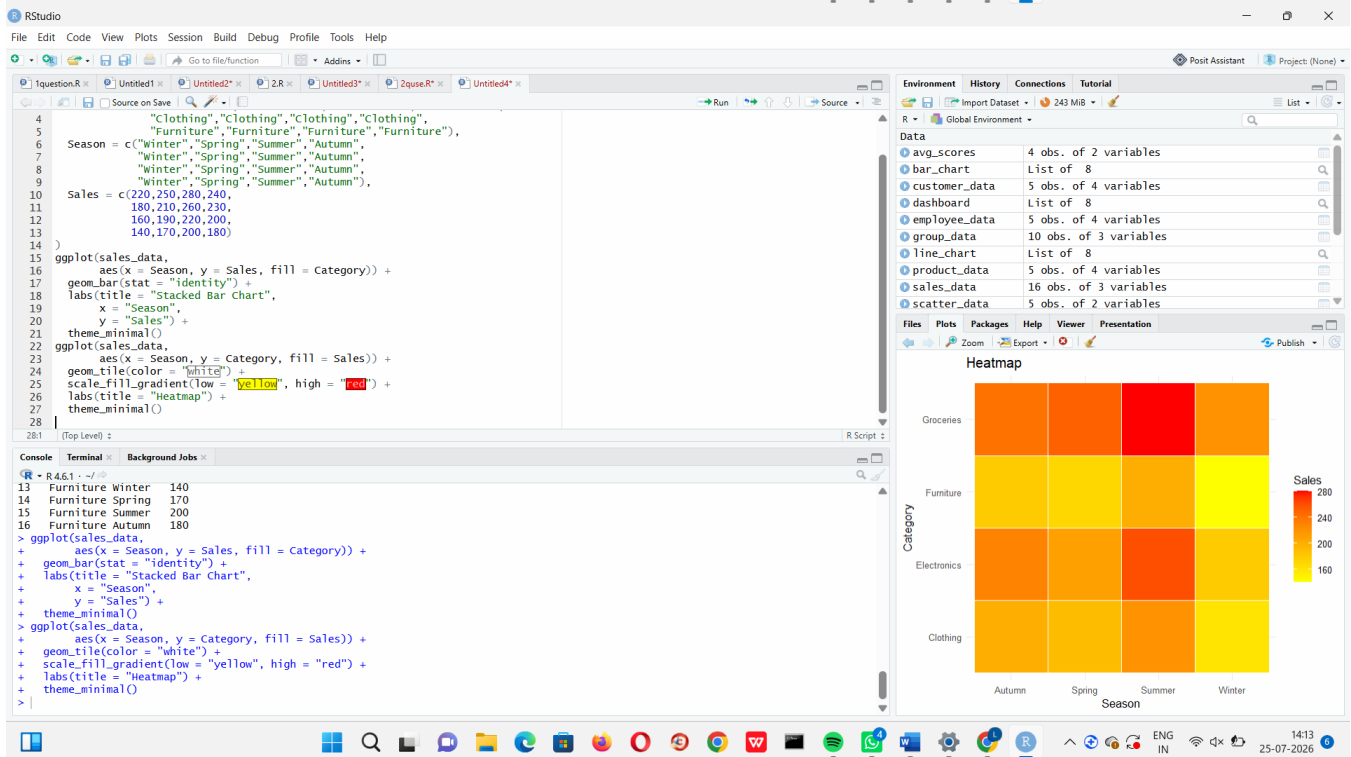
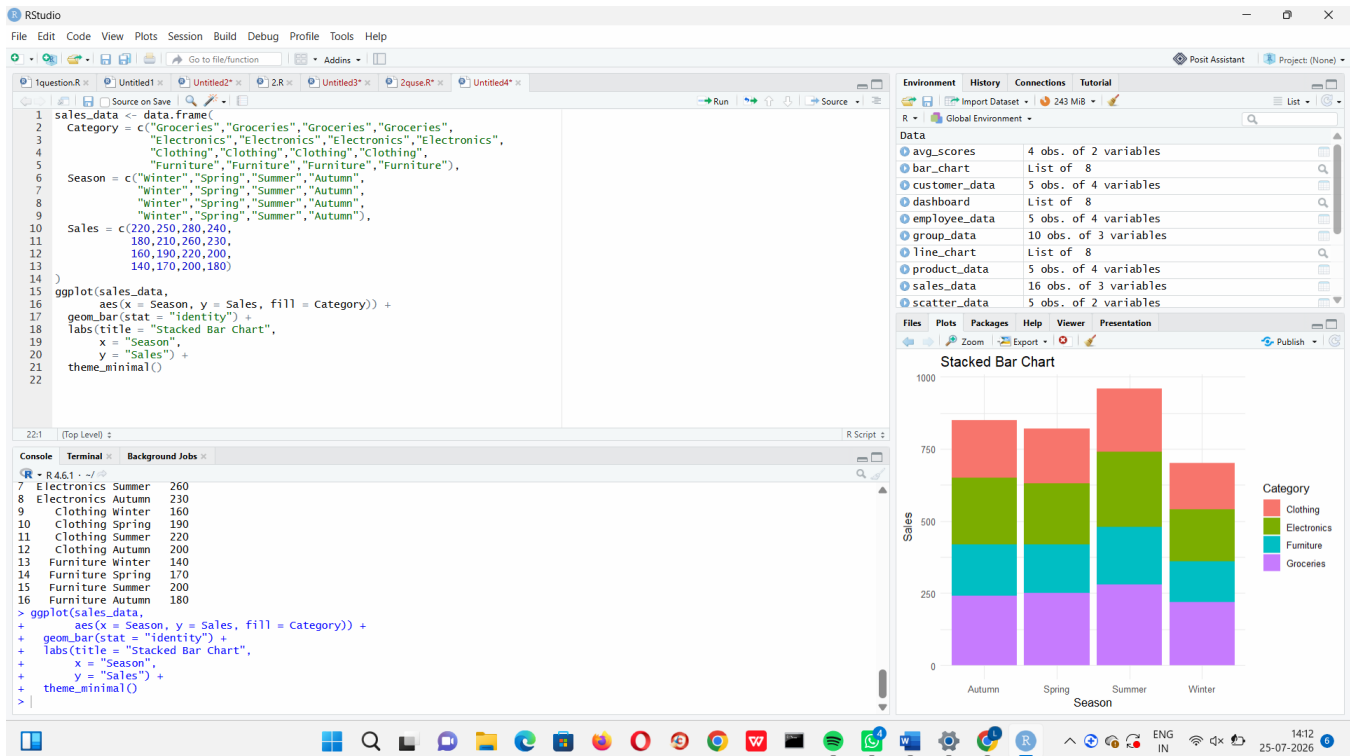


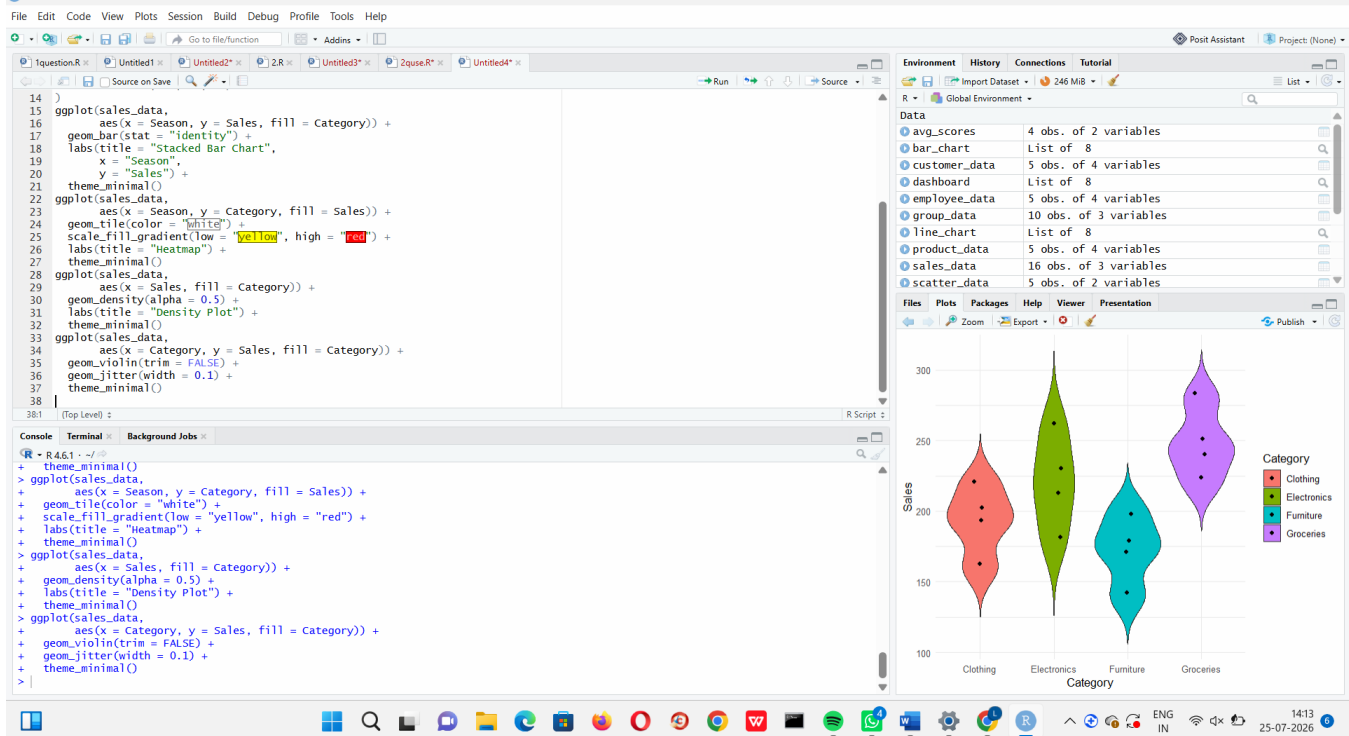
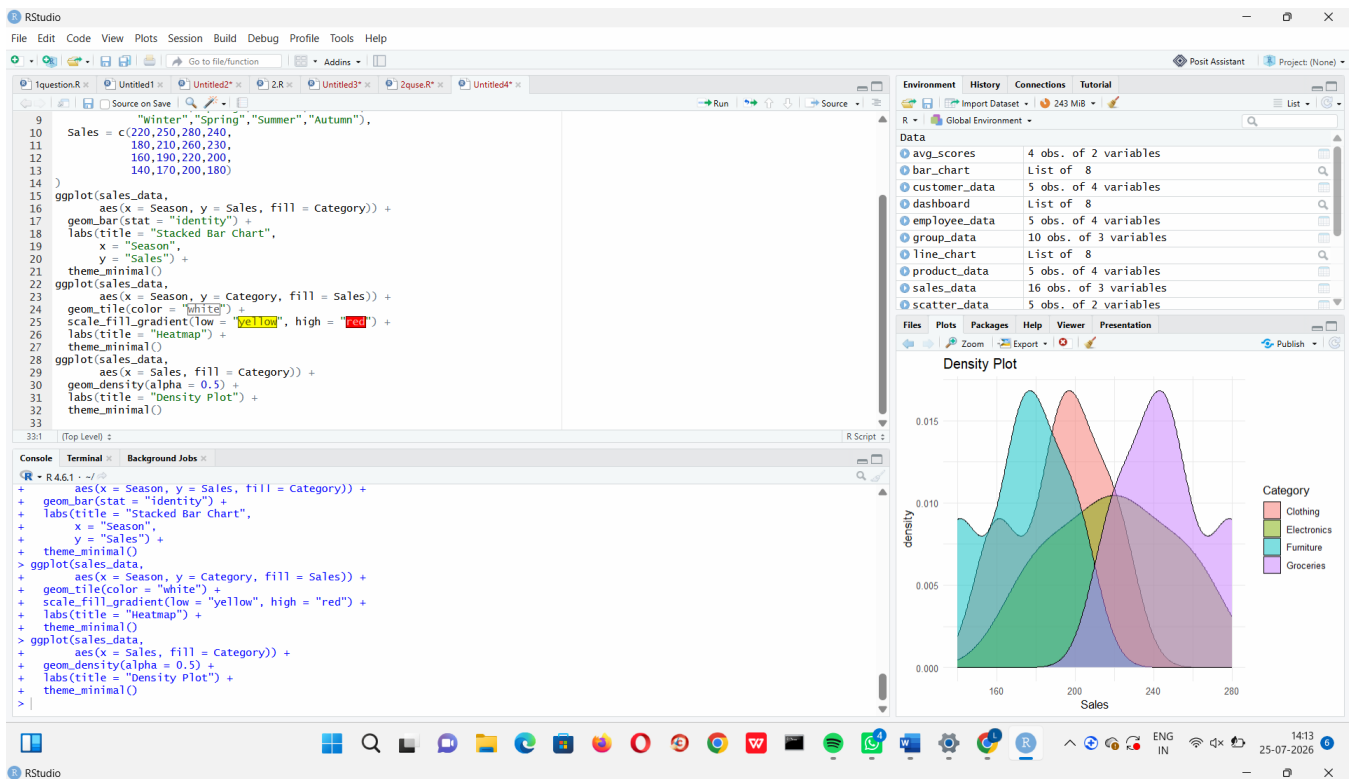




2.A supermarket chain collects data on customer purchases across various product categories and seasons. The management wants to understand spending patterns, compare sales distributions, and identify categories with the highest contribution to revenue.

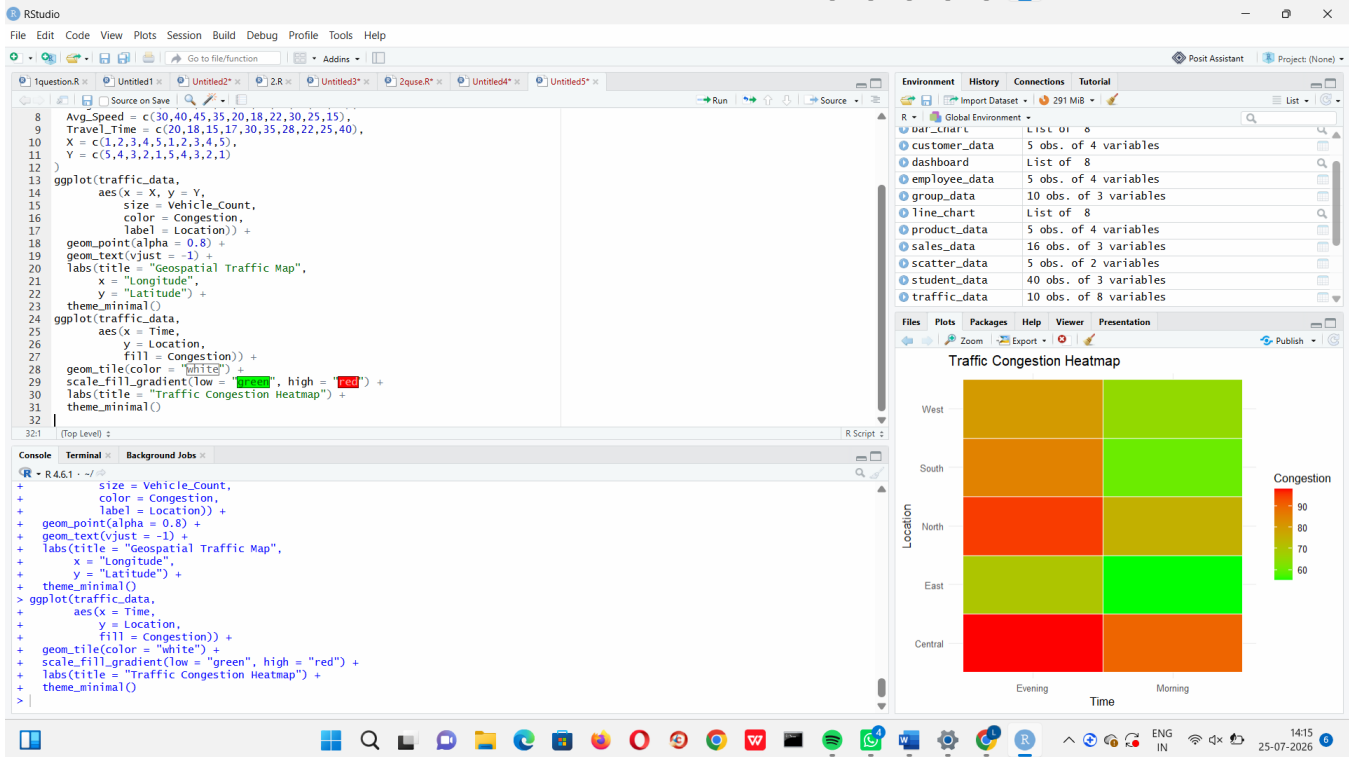
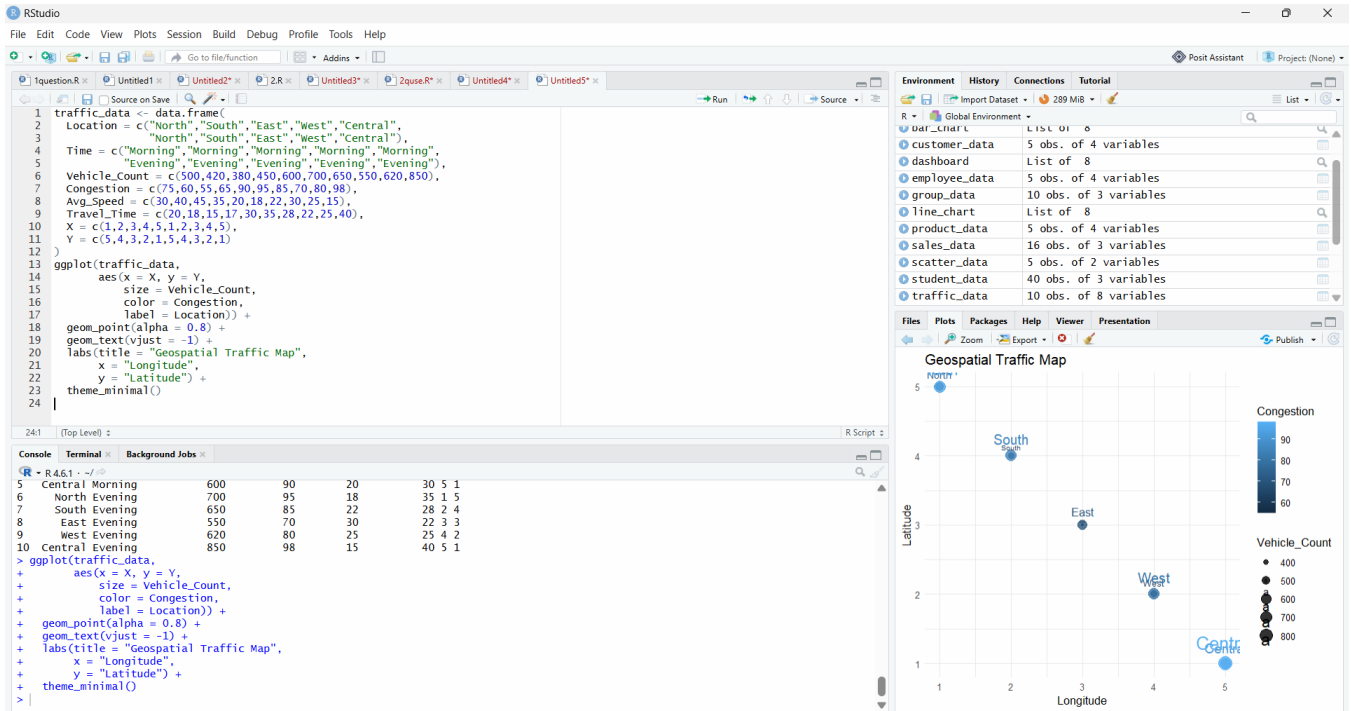
Task: Recommend appropriate visualization techniques such as stacked bars, heatmaps, density plots, and violin plots. Explain how these visualizations help reveal customer preferences, seasonal trends, and business opportunities.

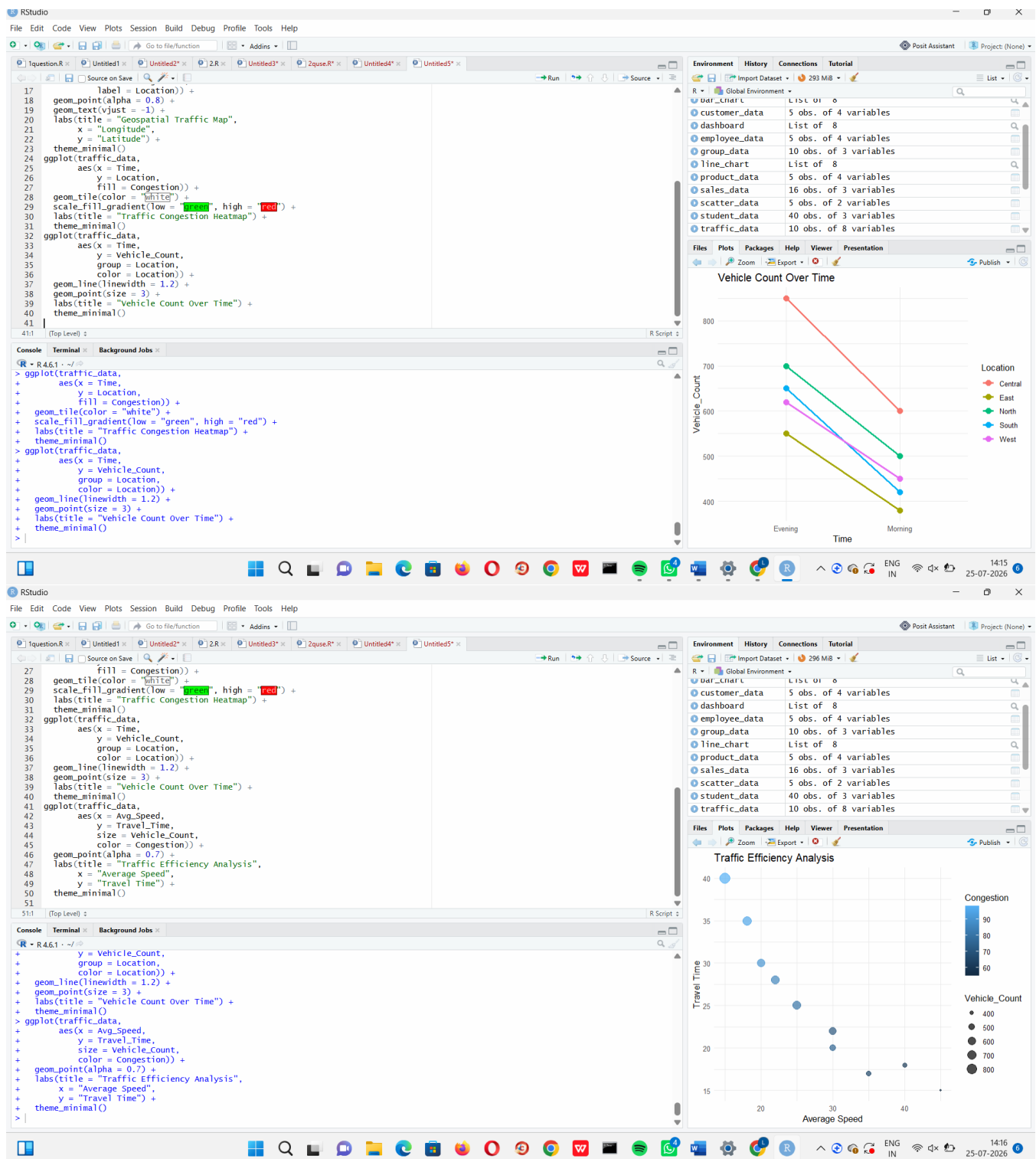




3.A smart city project gathers data on vehicle counts, traffic congestion levels, average speeds, and travel times across different locations and time periods. City planners aim to identify congestion hotspots and evaluate transportation efficiency.

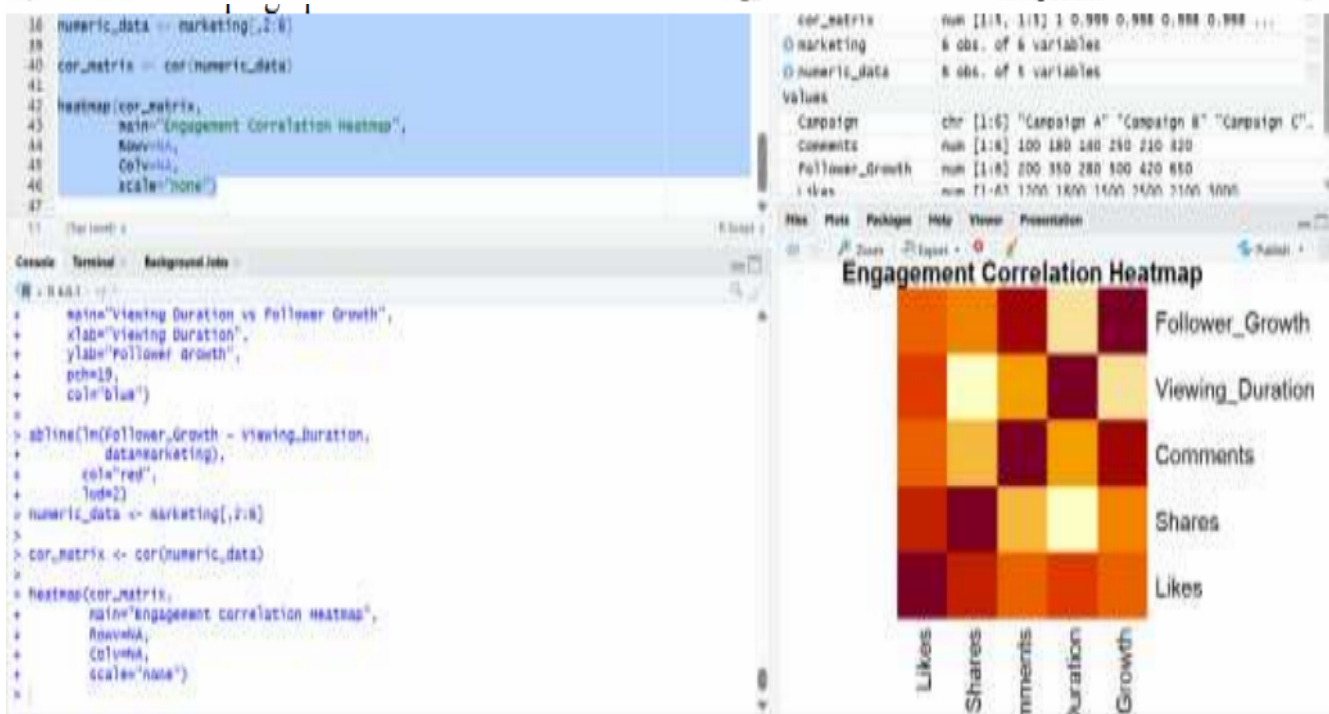
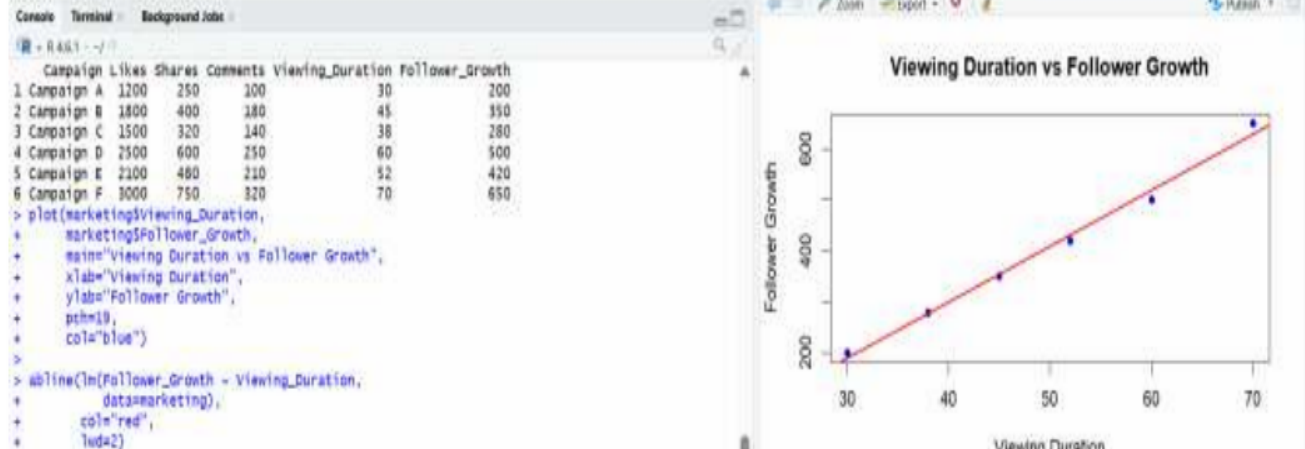
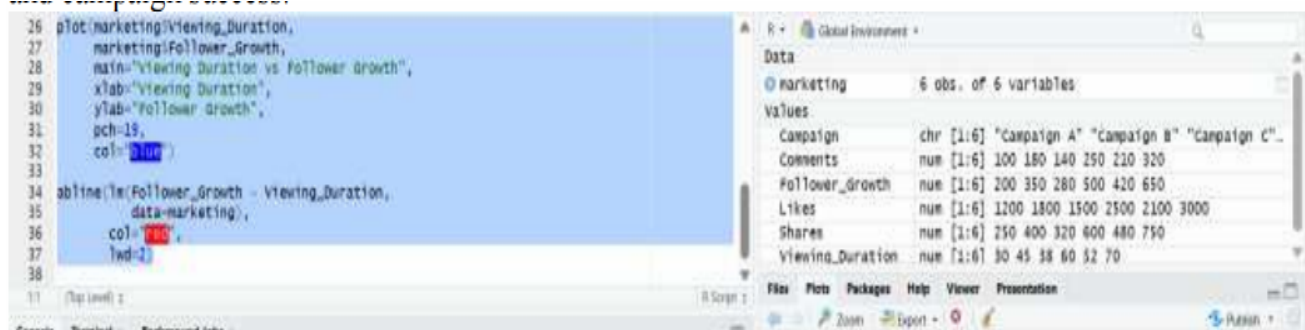
Task: Propose visualization methods including geospatial maps, heatmaps, temporal plots, and multivariate visualizations. Describe how these visualizations can assist in traffic management, route optimization, and infrastructure planning.



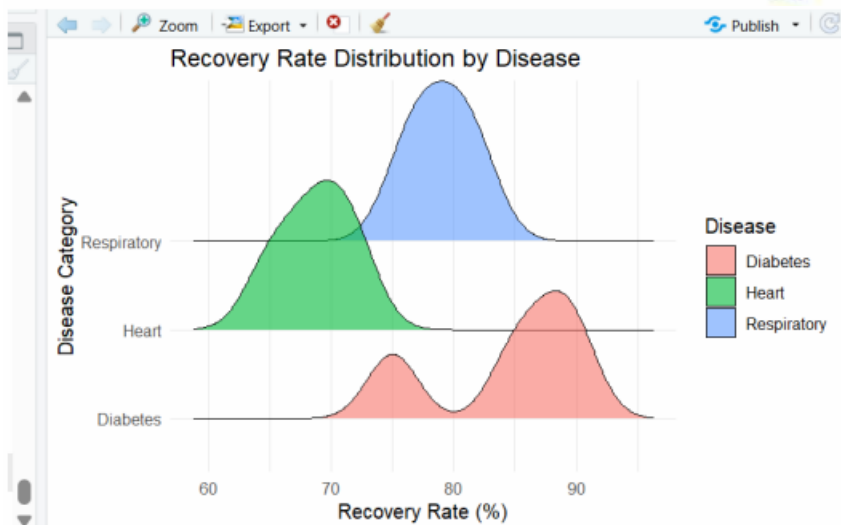
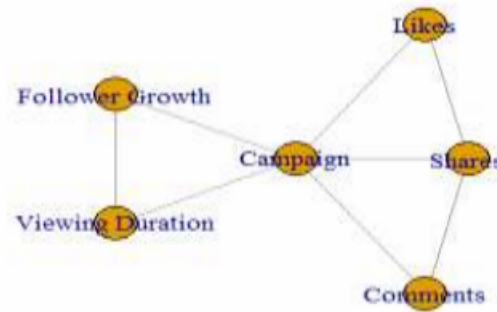


4.A digital marketing company tracks user engagement metrics such as likes, shares, comments, viewing duration, and follower growth across multiple campaigns. The goal is to analyze relationships between engagement variables and campaign success.

Task: Suggest suitable visualization techniques such as scatter plots, heatmaps, network visualizations, and multivariate plots. Explain how these visualizations help identify influential factors, audience behavior patterns, and effective marketing strategies.

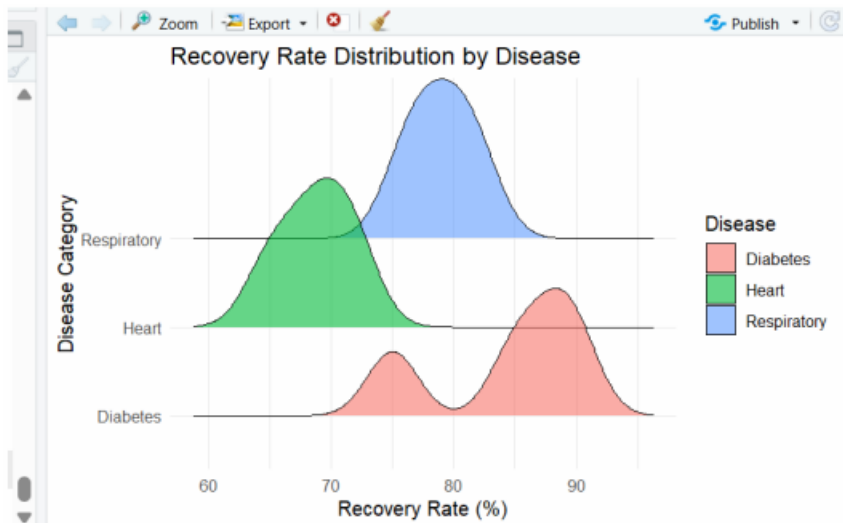


Marketing Engagement Network

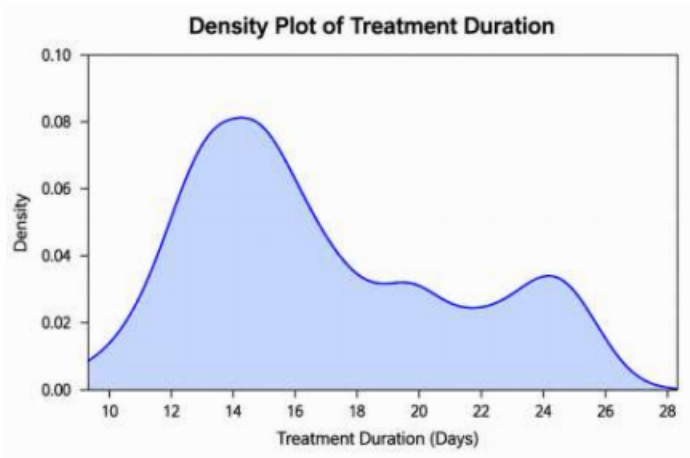


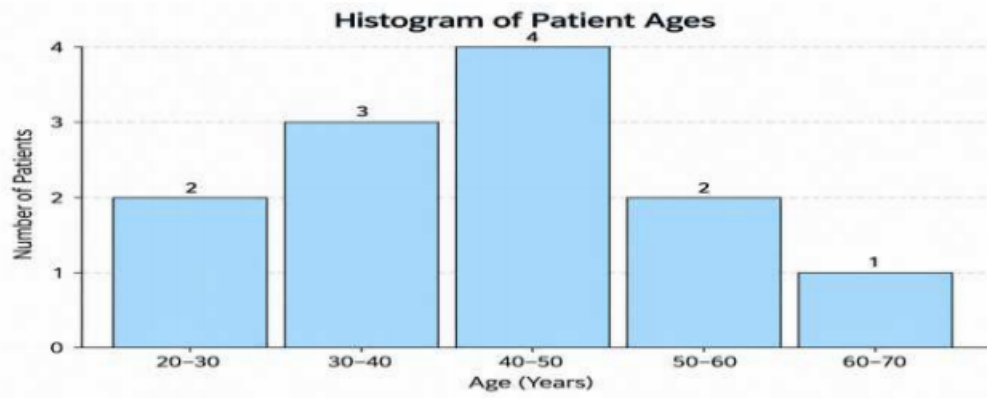
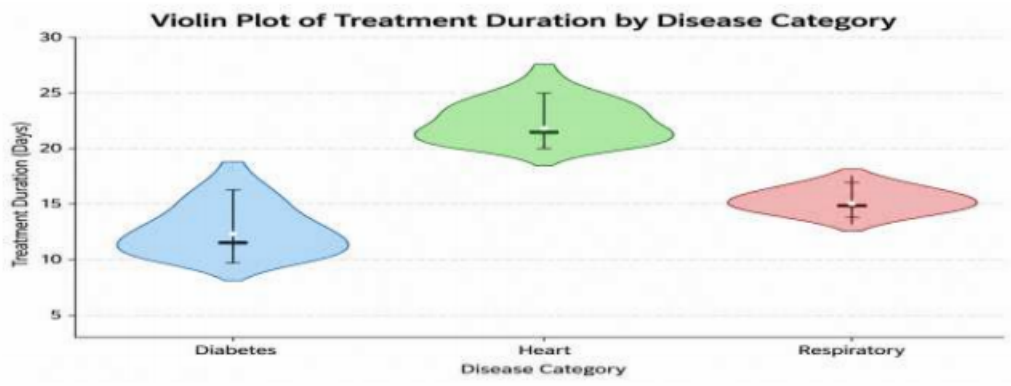
3. Density Plot

5.A healthcare organization maintains records of patients' ages, disease categories, treatment durations, recovery rates, and hospital visits. Medical administrators want to compare recovery distributions among patient groups and identify healthcare trends. Task: Select appropriate visualizations such as ridgeline plots, density plots, violin plots, histograms, and boxplots. Explain how these visualizations support disease analysis, treatment evaluation, and healthcare resource planning.



3 Density Plot





Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks	
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context		
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts		
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution		
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools		
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____